LABSHEET 3

A. Number Functions

1. Demonstrate ABS() with suitable examples.

```
SELECT ABS(-25) AS Result_1,
 ABS(17) AS Result_2,
 ABS(-8.75) AS Result_3;
```

2. Demonstrate ROUND() with suitable examples.

```
SELECT ROUND(12.345, 2) AS Two_Decimals,
 ROUND(12.345, 1) AS One_Decimal,
 ROUND(12.6) AS Rounded_Integer;
```

3. Demonstrate CEIL() and FLOOR().

```
SELECT CEIL(4.2) AS Ceiling_Value,
 FLOOR(4.8) AS Floor_Value,
 CEIL(-4.2) AS Ceiling_Negative,
 FLOOR(-4.2) AS Floor_Negative;
```

4. Demonstrate MOD() with suitable examples.

```
SELECT MOD(17, 5) AS Remainder_1,
 MOD(20, 4) AS Remainder_2,
 MOD(13, 2) AS Remainder_3;
```

5. Calculate the rounded salary of employees.

```
SELECT Emp_ID, Emp_Name,
 ROUND(Salary, 0) AS Rounded_Salary
FROM Employee;
```

B. Aggregate Functions

6. Find the total number of students using COUNT().

```
SELECT COUNT(*) AS Total_Students
FROM Student;
```

7. Find the total salary of all employees using SUM().

```
SELECT SUM(Salary) AS Total_Salary
FROM Employee;
```

8. Find the average salary using AVG().

```
SELECT AVG(Salary) AS Average_Salary
FROM Employee;
```

9. Find the highest salary using MAX().

```
SELECT MAX(Salary) AS Highest_Salary
FROM Employee;
```

10. Find the lowest salary using MIN().

```
SELECT MIN(Salary) AS Lowest_Salary
FROM Employee;
```

C. Character Functions

11. Demonstrate UPPER() on student names.

```
SELECT Name, UPPER(Name) AS Uppercase_Name
FROM Student;
```

12. Demonstrate LOWER() on student names.

```
SELECT Name, LOWER(Name) AS Lowercase_Name
FROM Student;
```

13. Demonstrate LENGTH() on student names.

```
SELECT Name, LENGTH(Name) AS Name_Length
FROM Student;
```

14. Demonstrate CONCAT() using first and last names.

```
SELECT CONCAT('Aman', ' ', 'Sharma') AS Full_Name;
```

```
SELECT CONCAT(First_Name, ' ', Last_Name) AS Full_Name
FROM Student;
```

15. Demonstrate SUBSTRING()/SUBSTR() with examples.

```
SELECT SUBSTRING('DATABASE', 1, 4) AS Part_1,
 SUBSTR('COMPUTER', 4, 4) AS Part_2;
```

D. Conversion Functions

16. Convert a number into a character/string value.

```
SELECT CAST(12345 AS CHAR) AS String_Value;
```

17. Convert a string into a numeric value.

```
SELECT CAST('456.78' AS DECIMAL(10,2)) AS Numeric_Value;
```

18. Convert a date value into a different format.

```
SELECT DATE_FORMAT('2026-10-11', '%d-%m-%Y') AS Formatted_Date;
```

19. Demonstrate CAST() with suitable examples.

```
SELECT CAST('2026-10-11' AS DATE) AS Date_Value,
 CAST(45.678 AS DECIMAL(5,2)) AS Decimal_Value,
 CAST(123 AS CHAR) AS Text_Value;
```

20. Demonstrate CONVERT() in MySQL.

```
SELECT CONVERT('789', UNSIGNED) AS Integer_Value,
 CONVERT('2026-10-11', DATE) AS Date_Value;
```

E. Date Functions

21. Display the current date using a suitable date function.

```
SELECT CURDATE() AS Current_Date;
```

22. Extract the year, month and day from a date.

```
SELECT YEAR('2026-10-11') AS Year_Value,
 MONTH('2026-10-11') AS Month_Value,
 DAY('2026-10-11') AS Day_Value;
```

23. Find the difference between two dates.

```
SELECT DATEDIFF('2026-10-11', '2026-10-01') AS Days_Difference;
```

24. Add a specified number of days to a date.

```
SELECT DATE_ADD('2026-10-11', INTERVAL 7 DAY) AS New_Date;
```

25. Display employees' joining dates in a required date format.

```
SELECT Emp_ID, Emp_Name,
 DATE_FORMAT(Joining_Date, '%d-%m-%Y') AS
Formatted_Joining_Date
FROM Employee;
```